MTH211 : Theory of Statistics

2025 - 26 (II)

Lecture 8: Sufficiency & Minimal Sufficiency

Some more examples on Sufficiency

Ex 6. $X_1, \ldots, X_n$ random sample from $N(\mu, \sigma^2)$ $\underset{\sim}{\theta} = (\mu, \sigma)' \in \Theta = \{(\mu, \sigma) : \sigma \in \mathbb{R}, \sigma > 0\}$

$$f_\theta(\underset{\sim}{x}) = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp. -\frac{1}{2\sigma^2} \sum_1^n (x_i - \mu)^2$$

$$= \frac{1}{(\sqrt{2\pi})^n} \frac{1}{\sigma^n} \exp\left\{-\frac{1}{2\sigma^2}\left(\sum x_i^2 - 2\mu \sum x_i + n\mu^2\right)\right\}$$

$$= h(\underset{\sim}{x}) \cdot g_{\underset{\sim}{\theta}}\left(\sum x_i, \sum x_i^2\right)$$

$T(\underset{\sim}{X}) = \left(\sum X_i, \sum X_i^2\right)$ is jointly sufficient for (μ, σ^2). (or, (μ, σ))

Ex 7 $X_1, \ldots, X_n$ r.s. from $U(\theta - 1/2, \theta + 1/2)$, $\theta \in \mathbb{R}$.

$$f_\theta(\underset{\sim}{x}) = \begin{cases} 1 & \text{if } \theta - 1/2 < x_1, \ldots, x_n < \theta + 1/2 \\ 0 & \text{ow} \end{cases}$$

$$= \begin{cases} 1 & \text{if } \theta - \frac{1}{2} < x_{(1)} < x_{(n)} < \theta + 1/2 \\ 0 & \text{ow} \end{cases}$$

$$= I_{(\theta - 1/2, x_{(1)})} I_{(x_{(n)}, \theta + 1/2)}$$

$T = (X_{(1)}, X_{(n)})$ jointly sufficient for θ.

Ex 8 $X_1, \ldots, X_n$ iid $N(\theta, \theta^2)$, $\theta > 0$

Find a sufficient statistic.

$\boxed{\text{Note}}$ (i) Any function of a sufficient statistic is not necessarily a sufficient statistic

($\text{Ex } 6$, $\sum X_i$ is a fn of $T(\underset{\sim}{X})$, but not sufficient for (μ, σ^2))

(ii) Any one-to-one function of a sufficient stat. is sufficient.

e.g., in Ex 6 $T(\underset{\sim}{X}) = (\bar{X}, S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2)$ is sufficient.

Prove this statement.

We are interested in a sufficient statistic that induces maximum reduction in data dimension. It is called a minimal sufficient statistic.

Defn Let $X_1, \ldots, X_n$ be a random sample from a distn P_θ, $\theta \in \Theta$ with jt. pmf/pdf $f_\theta(\underline{x})$. A statistic $T(\underline{X})$ is said to be minimal sufficient for θ if

(i) $T(\underline{X})$ is sufficient for θ, and

(ii) $T(\underline{X})$ is a function of every other sufficient statistic of θ.

(In Ex 3, $T_1(\underline{X}) = \sum X_i$, $T_2(\underline{X}) = \sum_1^{m} X_i + \sum_{m+1}^{n} X_i$, $T_3(\underline{X}) = X_1 + X_2 + \sum_3^n X_i$, ... all are sufficient stats, but only $T_1(\underline{X}) = \sum X_i$ is a function of $T_2, T_3, \ldots$ and is a MSS for θ).

The following result is for finding a minimal sufficient statistic.

Let $f_\theta(\underline{x})$ be the jt. pmf/pdf of $X_1, \ldots, X_n$ from a $P_\theta \in \mathcal{P}$, $\theta \in \Theta$, $\underline{x} \in \mathcal{X}$ ($\mathcal{P}$: family of distributions ch. by θ, Θ: parameter space, $\mathcal{X}$: sample space). Suppose $\exists$ a fn $T(\cdot)$ $\ni$ for every 2 sample points $\underline{x}, \underline{y} \in \mathcal{X}$, the ratio $\dfrac{f_\theta(\underline{x})}{f_\theta(\underline{y})}$ is independent of θ iff $T(\underline{x}) = T(\underline{y})$. Then $T(\underline{X})$ is a minimal sufficient statistic for θ (or, P_θ, or $\mathcal{P}$).

Ex 1 $X_1, \ldots, X_n$ r.s. from $N(\theta, 1)$, $\theta \in R$

for $\underline{x}, \underline{y} \in \mathcal{X}$,
$$\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = \frac{\left(\frac{1}{\sqrt{2\pi}}\right)^n \exp. -\frac{1}{2}\sum_1^n (x_i - \theta)^2}{\left(\frac{1}{\sqrt{2\pi}}\right)^n \exp -\frac{1}{2}\sum (y_i - \theta)^2} = \exp\left\{-\frac{1}{2}\left(\sum x_i^2 - \sum y_i^2\right)\right\} \exp\left\{\theta\left(\sum x_i - \sum y_i\right)\right\}$$
is indep. of θ iff $\sum x_i = \sum y_i$ $\Rightarrow T(\underline{X}) = \sum_1^n X_i$ is a MSS for θ.

Ex 2. $X_1, \ldots, X_n$ r.s. from $N(\mu, \sigma^2)$, $\underline{\theta} = (\mu, \sigma)'$, $\mu \in R$, $\sigma > 0$.

for $\underline{x}, \underline{y} \in \mathcal{X}$,
$$\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = \exp.\left\{-\frac{1}{2}\left(\frac{1}{\theta^2}\sum x_i^2 - \sum y_i^2\right) - \frac{2}{\theta}\left(\sum x_i - \sum y_i\right)\right\}.$$
is indep. of θ iff $\sum x_i = \sum y_i$ and $\sum x_i^2 = \sum y_i^2$.

$\therefore T(\underline{X}) = (\sum X_i, \sum X_i^2)$ jtly minimal sufficient ~~for~~ for (μ, σ^2)

Ex 3. $X_1, \ldots, X_n$ r.s. from $U(0, \theta)$, $\theta > 0$.

for $\underline{x}, \underline{y} \in \mathcal{X}$,
$$\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = \frac{\frac{1}{\theta^n} I_{(x_{(n)}, \theta)} I_{(0, x_{(1)})}}{\frac{1}{\theta^n} I_{(y_{(n)}, \theta)} I_{(0, y_{(1)})}}$$
independent of θ iff $x_{(n)} = y_{(n)}$

$\Rightarrow T(\underline{X}) = X_{(n)}$ MSS for θ

Next Unbiased Estimator.